

1.INTRODUCTION

The Periyar University NSS Portal is a web-based application designed to digitize and simplify the management of National Service Scheme (NSS) activities across different institutional levels. In the existing system, processes such as volunteer enrollment, event management, and report submission are carried out manually, leading to inefficiency, errors, and poor data management. This project introduces a centralized platform that connects administrators, colleges, and NSS units, ensuring better coordination and transparency.

The system is structured into multiple functional modules including Authentication & Secure Access, Administrator Command, College Administration, Unit Operations & Volunteer Management, Student Enrolment & Demographic, Automated Document & Reporting, and Event Management & Field Activity modules. Each module is designed to handle specific operations, ensuring a well-organized and scalable system architecture.

The system follows a role-based access structure, where Admin, College, and Unit users have specific permissions and responsibilities. It is developed using modern technologies to ensure performance and scalability. The frontend is built using React with Vite for a fast and responsive interface, while the backend uses Node.js and Express to handle server-side operations. MongoDB is used for efficient data storage, and communication between components is handled through RESTful APIs. Secure authentication is implemented to protect user data, along with additional features like email notifications.

The application provides functionalities such as digital volunteer enrollment, event creation and tracking, collaborative participation, and report generation. Overall, the system improves efficiency, reduces manual work, and offers a reliable solution for managing NSS activities in a structured and effective manner.

1.2.SYSTEM SPECIFICATION :

1.2.1.HARDWARE CONFIGURATION:

- **Processor** : Dual-Core Processor or higher (e.g., Intel Core i3 or equivalent) with a clock speed of 2.0 GHz or higher.
- **RAM** : Minimum 4 GB of RAM.
- **Storage** : Minimum 512 MB of free disk space.
- **Network** : Stable internet connection for cloud database connection(MongoDB atlas)

The selected hardware configuration provides sufficient performance to run development tools like Node.js, React, and database services smoothly. A minimum of 4 GB RAM and a dual-core processor ensures efficient multitasking during coding and testing. Stable internet connectivity is essential for accessing cloud services like MongoDB Atlas and Cloudinary.

1.1.2.SOFTWARE SPECIFICATION:

- **Operating System** : Windows 10 or 11, Linux (Ubuntu 16.04 or later).
- **Framework** : Node.js with Express.js for backend.
- **Libraries** : React.js for frontend.
- **Package Manager** : Node Package Manager (npm) for managing dependencies.
- **Database** : MongoDB (Atlas for Cloud Database).
- **API Testing** : Postman.
- **Cloud storage** : Cloudinary (For storing event brochures)
- **Version Control** : Git and GitHub.
- **Code Editor** : Visual Studio Code (or) CodeSandbox(for online coding).
- **Browser** : Latest versions of Chrome, Firefox, Microsoft Edge or Brave.

The chosen MERN stack enables efficient full-stack development using a single language (JavaScript), improving development speed and consistency. Node.js and Express.js support scalable backend development, while React.js provides a dynamic user interface. Additionally, MERN applications can be easily deployed on cloud platforms, making the system flexible and accessible. Tools like MongoDB Atlas, GitHub, and Postman ensure reliable database management, version control, and API testing.

2.SYSTEM STUDY

2.1.EXISTING SYSTEM

2.1.1. DESCRIPTION

The existing system for managing NSS activities is mostly manual and paper-based. Processes like volunteer enrollment, event management, and report submission are handled using physical forms such as Annexure-A. Data is stored in files or registers, making it difficult to organize and retrieve information.

In addition, the Existing NSS webpage <https://www.periyaruniversity.ac.in/nss.php> mainly provides downloadable links for forms and displays a few past event images, but lacks interactive features or a centralized digital system for managing activities. Communication between administrators, colleges, and units is not centralized, leading to delays and confusion.

2.1.2. LIMITATIONS

- **Manual Process:** Most NSS activities such as enrollment, attendance, and report preparation are handled using paper-based methods, increasing manual effort.
- **Time Consumption:** The overall process is slow, as tasks like data entry, verification, and report generation require significant time.
- **Human Errors:** Manual handling of data often leads to mistakes in recording, duplication, and loss of important information.
- **Poor Data Management:** Maintaining records in physical files or basic formats makes it difficult to store, organize, and retrieve data efficiently.
- **Lack of Centralization:** There is no single unified platform to manage all activities, leading to scattered and uncoordinated operations.
- **Low Transparency:** Administrators and higher authorities have limited visibility into the activities of colleges and units.
- **Tracking Issues:** Monitoring volunteer participation, event progress, and outcomes becomes challenging without a proper system.

2.2.PROPOSED SYSTEM

2.2.1. DESCRIPTION

The proposed system is a web-based **Periyar University NSS Portal** designed to automate and simplify NSS activities through a centralized platform. It replaces manual processes like volunteer enrollment, event management, and reporting with a digital solution. The system uses role-based access control for Admin, College, and Unit users to ensure secure and organized operations. It allows easy storage, retrieval, and management of data in real time. The platform also supports collaborative events and improves communication between different levels. Automated notifications and structured report generation enhance efficiency. Overall, the system reduces manual work, minimizes errors, and improves transparency.

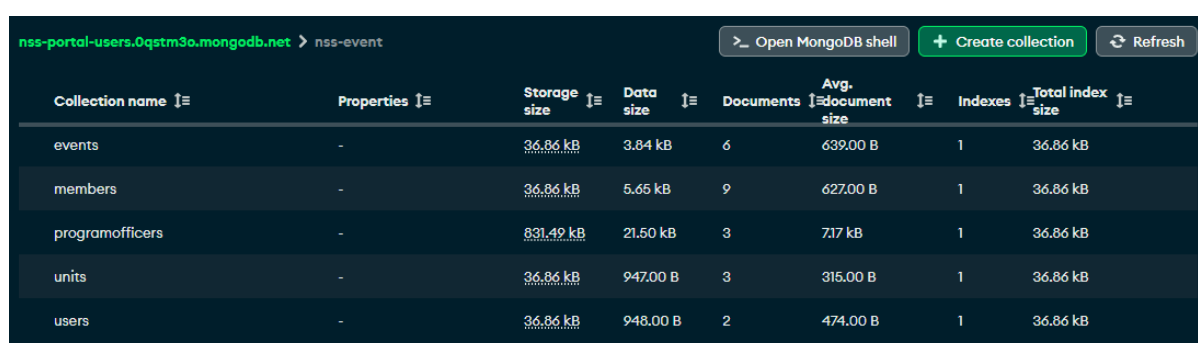
2.2.2. FEATURES

- **Digital Enrollment:** Provides an online system for registering NSS volunteers with detailed personal, and activity-related information, replacing paper-based forms.
- **Role-Based Access Control:** Ensures secure and structured access for different users such as Admin, College, and Unit, with specific permissions for each role.
- **Event Management:** Enables users to create, update, and manage NSS events, including scheduling, descriptions, and tracking progress.
- **Collaborative Events:** Allows multiple NSS units to participate and coordinate in a single event, improving teamwork and resource sharing.
- **Centralized Data Storage:** Maintains all data in a unified database, making it easy to access, update, and manage information efficiently.
- **Automated Notifications:** Sends real-time alerts and updates to users regarding new events, changes, and report submissions.
- **Report Generation:** Automatically generates structured reports for events and volunteer participation, reducing manual effort.
- **Data Export Options:** Supports exporting data in formats such as PDF, Excel, and CSV for administrative and official purposes.

3. SYSTEM DESIGN AND DEVELOPMENT

3.1.DATABASE DESIGN

The database design of the **Periyar University NSS Portal** is structured to efficiently manage data related to users, units, volunteers, events, and program officers. It follows a flexible schema approach to handle different types of data while maintaining proper relationships between entities. Each component in the system is interconnected, ensuring smooth data flow and consistency. The design supports scalability, allowing the system to handle increasing data over time. It also ensures efficient storage, retrieval, and updating of records.



The screenshot shows the MongoDB Atlas interface for the database 'nss-portal-users.0qstm3o.mongodb.net' and the 'nss-event' database. At the top, there are buttons for 'Open MongoDB shell', 'Create collection', and 'Refresh'. Below these is a table listing the collections in the database.

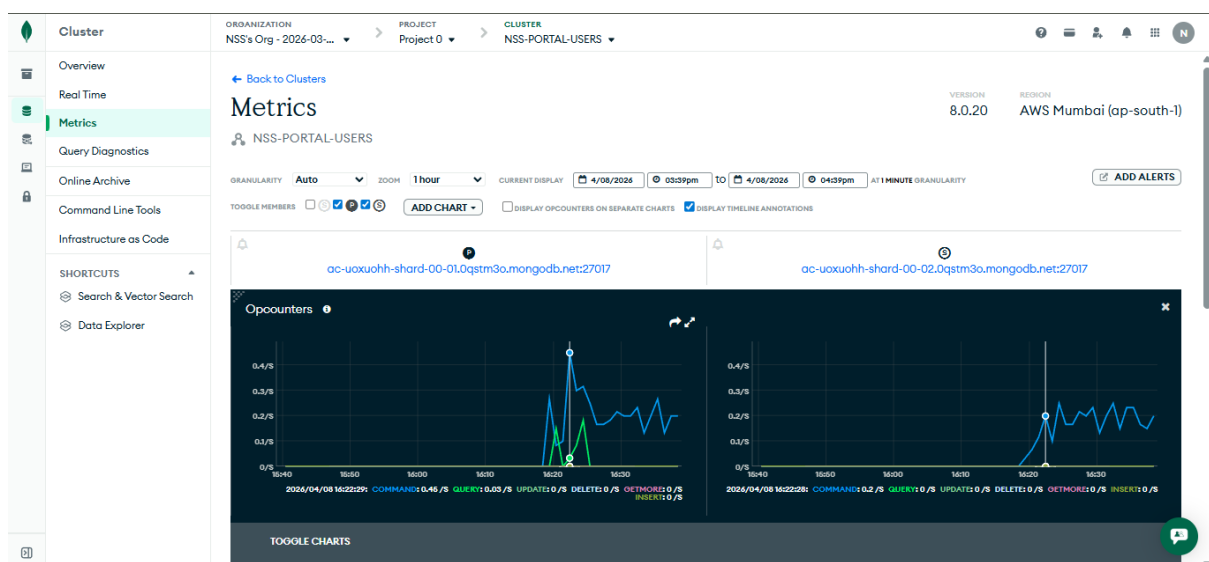
Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
events	-	36.86 kB	3.84 kB	6	639.00 B	1	36.86 kB
members	-	36.86 kB	5.65 kB	9	627.00 B	1	36.86 kB
programofficers	-	831.49 kB	21.50 kB	3	717 kB	1	36.86 kB
units	-	36.86 kB	947.00 B	3	315.00 B	1	36.86 kB
users	-	36.86 kB	948.00 B	2	474.00 B	1	36.86 kB

The **Member**, **Unit**, and **User** models form the core structure of the system. The Member model stores detailed volunteer information, including personal, academic, and enrollment details, and is linked to specific units and colleges. The Unit model manages NSS units, their members, and associated activities within a college. The User model represents institutional-level data, including login credentials and organizational details, and maintains relationships with multiple units.

The **Event** and **Program Officer** models handle operational and administrative functionalities. The Event model stores complete event details such as schedule, venue, participants, collaborators, and final reports. It also supports tracking attendance and post-event outcomes. The Program Officer model maintains details of NSS coordinators, including their personal and professional information. Together, these models ensure efficient data management, reduce redundancy, and support the overall functioning of the system.

Table(Collection) Name	Purpose
User (College)	Manages institution login, overall units, events, and adopted villages
AdoptingVillages (Sub-table)	Stores details of villages adopted by the college for NSS activities
Unit	Represents an NSS unit under a college, manages members and events
Member	Stores student details including personal info and NSS enrolment data
Event	Manages event details like schedule, venue, participants, and unit association
ProgramOfficer	Stores details of NSS officers who manage and guide units
Collaborators (Sub-table)	Handles collaboration between different units for an event
Report (Sub-table)	Stores post-event report details like participants count, outcome, and files

MongoDB is used as the database management system to store and manage all application data efficiently. It provides a flexible, document-based structure that allows easy handling of complex and dynamic data. As we use the MongoDB Atlas (Cloud) Its scalability and high performance support the smooth functioning of the Periyar University NSS Portal .



3.2.INPUT DESIGN

Basic Information

EVENT NAME *

Enter event name

EVENT DESCRIPTION *

EVENT CATEGORY *

Select Category

☒

THIS IS A SINGLE-DAY EVENT

DATE *

mm/dd/yyyy

TIME FROM

--:-- --

TIME TO

--:-- --

VENUE *

Enter event venue

Media & Documents

EVENT IMAGES

Choose Files

No file chosen

Upload Selected Images

No images selected or uploaded yet.

EVENT BROCHURE (PDF)

Choose File

No file chosen

Upload Brochure

Collaboration

COLLABORATING UNITS (OPTIONAL)

Select a Unit to Invite

Add

Selected units will receive an invitation to collaborate on this event.

Cancel

Create Event

7

Register Program Officer Pro-form-a

×

Main Details

NAME OF PROGRAM OFFICER

DESIGNATION

OFFICER-4

Professor

DEPARTMENT

UNIT (OPTIONAL)

Computer Science

Unassigned

Personal Details

PROFILE IMAGE

CURRENT PHOTO

Choose File

sample-profile.png



Download Form PDF

Close

Register Officer

The input design of the **Periyar University NSS Portal** focuses on providing a user-friendly and consistent interface for data entry. All input fields such as text boxes, dropdowns, date pickers, and file uploads are designed with a clean and modern style using rounded corners, proper spacing, and readable fonts. Interactive states like hover and focus are included to guide users during data entry. The system also applies input validation to ensure accuracy and reduce errors. This improves overall usability and ensures reliable data collection.

The application uses structured forms for different modules to simplify complex inputs. The **Event Creation Form** allows users to enter details like event name, category, date, venue, and collaborators in an organized manner. The **Volunteer Enrollment Form (Annexure-A)** is divided into sections such as personal, academic, physical, and skills to reduce user complexity. Advanced UI elements like icons, placeholders, and grouped inputs further enhance clarity. Overall, the input design ensures efficient, accurate, and smooth user interaction.

3.3.OUTPUT DESIGN



Registered Colleges2 total

Click a row to enter the college dashboard

CODE	INSTITUTION NAME	UNITS	EVENTS	ACTIONS
888	Sample Institute	2	5	Remove
333	Sample2	1	1	Remove

Admin DashboardSYSTEM

Periyar University NSS Portal

+ Register CollegeSign Out

Events CalendarView All --

April 2026

Sun	Mon	Tue	Wed	Thu	Fri	Sat
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

Click on dates with colored dots to view events

Quick Actions

Manage All Events

Program Officers

Student Records

Total Events6

Upcoming Events3

Address: 9/197 Udhayam Nagar, Kamalapuram, Omalur Salem

ACADEMIC & ETI

Educational Qualification: MCA

ETI Training Completed: Yes

Seminars / Workshops / Courses:

• NIL

11	MOBILE EMAIL ID NO.	9988 / 10002
12	E-mail ID	dinesh.m2003@gmail.com
13	Aadhaar Card No.	456789123456
14	Date/Year of Enrolment	2026-03-27
15	Cultural Talents (Please specify)	Folk Dance
16	Hobbies	Football, Photography

Declaration

I DineshKumarM hereby agree to obey all the rules and regulations of National Service Scheme and work imbibing the spirit of the Scheme.

GENERAL

The output design of the **Periyar University NSS Portal** focuses on presenting data in a clear, structured, and visually appealing manner. The system provides dashboards with graphical representations such as charts and key metrics to help administrators easily analyze NSS activities. Visual elements like charts and calendars are used to display event distribution, institutional performance, and daily schedules. These outputs improve decision-making by providing real-time insights and better understanding of data.

The system also supports professional reporting and data export functionalities. It generates detailed PDF reports for completed events, including participation details, outcomes, and supporting images. Additionally, data can be exported in formats like Excel for offline use and official submissions. Automated email notifications are used to communicate updates and alerts to users. The system also produces standardized documents such as the Annexure-A enrollment form. Overall, the output design ensures effective data presentation, reporting, and communication.

3.4.CODE DESIGN

The code design of the **Periyar University NSS Portal** follows a well-structured and modular architecture based on modern software engineering principles. The application is developed using the MERN stack, ensuring scalability, flexibility, and efficient handling of dynamic data. The frontend is built using a component-based approach, where reusable components such as forms, modals, dashboards, and input elements improve code reusability and maintainability. State management is handled using Context API, enabling smooth data flow across components, while routing is implemented with proper protection mechanisms to restrict unauthorized access.

The backend is designed following RESTful API architecture. Each API endpoint is structured with proper request handling and standardized responses, along with effective error handling using try-catch mechanisms. Security is a key aspect of the design, implemented through JWT-based authentication and role-based access control to manage Admin, College, and Unit privileges. Sensitive data such as passwords are securely handled using encryption techniques. The system also integrates third-party services for functionalities like email notifications, media storage, and data visualization. Overall, the code design ensures a secure, scalable, and maintainable system that supports efficient development and future enhancements.

frontend/

```
|— src/
|   |— components/    # All UI modules & Dashboards
|   |— context/       # State management (User/Theme)
|   |— assets/        # Global images & icons
|   └— App.jsx        # Main Routing & Entry point
```

backend/

```
|— server.js          # Core server logic & API Routes
|— middleware/        # Auth & Security checks
└— .env               # Secure configuration
```

3.5.SYSTEM DEVELOPMENT - (Agile Model)

Requirement Analysis

This phase involves understanding the needs and objectives of the NSS Portal. The system requirements are gathered by identifying the functionalities such as member management, unit handling, and activity tracking. Both functional and non-functional requirements are clearly defined to ensure the system meets user expectations.

System Design

In this phase, the overall architecture of the system is planned. The application is designed using the MERN stack, ensuring a clear separation between frontend and backend. Database schema, user interface layout, and system workflows are structured to provide efficiency, scalability, and ease of use.

Implementation (Development)

During implementation, the actual coding of the system is carried out. The frontend is developed using React.js to create an interactive user interface, while the backend is built using Node.js and Express.js to handle server-side operations. MongoDB is used to store and manage data efficiently.

Testing

The system is thoroughly tested to ensure all functionalities work correctly. Different types of testing such as unit testing, integration testing, and user testing are performed. Errors and bugs are identified and fixed to improve system reliability and performance.

Deployment

After successful testing, the system is deployed to a production environment (netlify for frontend and render for backend). This makes the NSS Portal accessible to users. Proper configuration and hosting are ensured so that the application runs smoothly in real-time usage.

Maintenance

Maintenance is an ongoing process where the system is monitored and updated regularly. Any issues faced by users are resolved, and new features or improvements are added to enhance performance and usability over time.

3.5.1. MODULE DESCRIPTION

1. Authentication & Secure Access Module

- **Role-Based Access Control (RBAC):** Implements specialized login gateways for three distinct user levels: System Administrator, College Coordinator, and NSS Unit Head.
- **JWT Security Integration:** Ensures data integrity using JSON Web Tokens (JWT) for all API requests, providing end-to-end encryption for session management.
- **Aesthetic UI Persistence:** Includes a specialized Theme Toggle system that dynamically switches between Light and Night modes with state persistence.

The image displays three distinct login interfaces side-by-side, each with a light blue header and a white body. The first interface on the left is for the 'SYSTEM ADMINISTRATOR' and features a blue icon, the title 'Admin Sign In', and the subtitle 'Periyar University NSS Portal'. It includes input fields for 'USERNAME' (with placeholder 'Enter admin username') and 'PASSWORD' (with placeholder 'Enter password' and a toggle icon), a blue 'Sign In' button, and a 'Back to Home' link. The middle interface is for the 'NSS UNIT' and features a green icon, the title 'Unit Sign In', and the subtitle 'Access your NSS unit dashboard'. It includes input fields for 'COLLEGE CODE' (with placeholder 'e.g. 507'), 'UNIT CODE' (with placeholder 'e.g. PUNSS...'), and 'PASSWORD' (with placeholder 'Enter unit password' and a toggle icon), a blue 'Sign In' button, and a 'Back to Home' link. The third interface on the right is for the 'ORGANIZATION' and features a blue icon, the title 'Organization Sign In', and the subtitle 'Access your college NSS dashboard'. It includes input fields for 'USERNAME' (with placeholder 'Enter your username') and 'PASSWORD' (with placeholder 'Enter your password' and a toggle icon), a blue 'Sign In' button, and a 'Back to Home' link.

2. Administrator Command Module (Centralized Oversight)

- **Institutional Monitoring:** Provides a bird's-eye view of all registered colleges, their organizational structures and the events.
- **Master Student Registry:** A searchable database of all students across the region, featuring multi-layered filtering by community, college, and department.
- **Officer Management:** Centralized tracking and management of Program Officers' registration and performance metrics.

System Administrator

All Events Management

[Back to Dashboard](#)

SEARCH EVENTS

Search by name, code, or college...

FILTER BY CATEGORY

All Categories

Total Events

6

Filtered Results

6

Categories

6

Events List

EVENT CODE	EVENT NAME	CATEGORY	COLLEGE	UNIT	DATE(S)	VENUE	ACTIONS
NSSEVTPUNSSAM88801001	Sample Event	Tree Plantation	Sample Institute 888	unit1 PUNSSAM88801	2026-04-08	Periyar university Campus	View Details
NSSEVTPUNSSAM33301001	Sample Event-2	Awareness Program	Sample2 333	unit2 PUNSSAM33301	2026-04-07	Seminar Hall, Dept of Computer Science, Periyar University	View Details
NSSEVTPUNSSAM88801002	Sample Event-3	Cleanliness Drive	Sample Institute 888	unit1 PUNSSAM88801	2026-04-09	University Campus	View Details
NSSEVTPUNSSAM88802001	Sample Event	Seminar	Sample Institute 888	Unit 2 PUNSSAM88802	2026-03-25	Seminar Hall, Dept of Computer Science, Periyar University	View Details
NSSEVTPUNSSAM88801003	event-test	Sports	Sample Institute 888	unit1 PUNSSAM88801	2026-04-10	Ground	View Details
NSSEVTPUNSSAM88801004	mail test	Workshop	Sample Institute 888	unit1 PUNSSAM88801	2026-04-12	seminar hall	View Details

All Program Officers

Admin Access

[Dashboard](#)

SEARCH BY NAME

Enter officer name...

FILTER BY COLLEGE

College name or code...

ID	OFFICER NAME	COLLEGE (CODE)	UNIT	DEPARTMENT	CONTACT	ACTION
NSSPO88803	Officer 3	Sample Institute 888	Unassigned	CS	9876543210	View Edit
N/A	officer 3	Sample Institute 888	PUNSSAM88801	Chemistry	9876543210	View Edit
N/A	Officer 2	Sample2 333	PUNSSAM33301	CS	9876543210	View Edit

All Students

Admin Access

[Dashboard](#)

SEARCH (NAME OR REG NO)

Enter student name or reg no...

FILTER BY COLLEGE

College name or code...

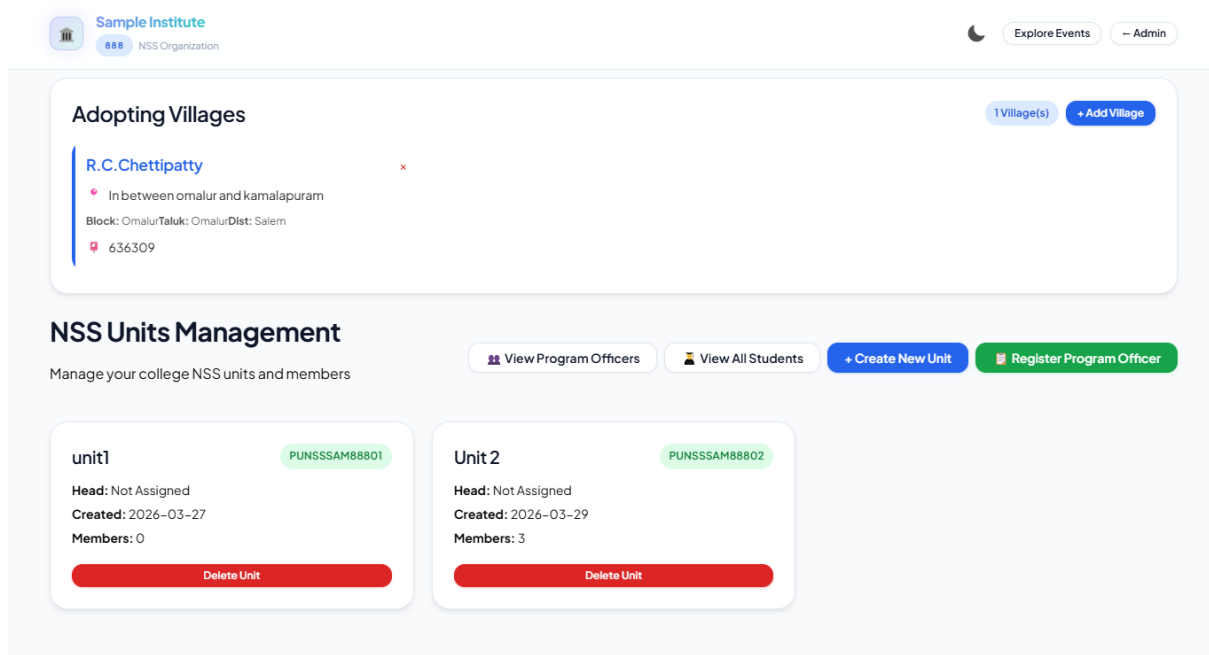
FILTER BY COMMUNITY

All Communities

REG NO	NAME	COLLEGE (CODE)	UNIT	DEPT	COMMUNITY	BATCH	ACTION
22EC045	PriyaS	Sample Institute 888	PUNSSAM88802	MCA	SC	2022 - 2026	✓ View Enrolment
22ME012	DineshKumarM	Sample Institute 888	PUNSSAM88802	MBA	MBC	2022 - 2026	✓ View Enrolment
22CS001	KarthikeyanR	Sample Institute 888	PUNSSAM88802	CS	OBC	2022 - 2026	✓ View Enrolment

3. College Administration Module (Institutional Dashboard)

- **NSS Unit Orchestration:** Facilitates the creation and management of up to six NSS units per institution.
- **Program Officer Recruitment:** Manages the appointment of officers, ensuring unique leadership per unit through intelligent synchronization logic.
- **Village Adoption System:** A dedicated interface for managing rural outreach data, capturing location-specific details like Block, Taluk, and District for adopted villages.



4. Unit Operations & Volunteer Management Module

- **Local Unit Control:** Enables Unit Heads to manage their specific group of assigned volunteers independently.
- **Volunteer Lifecycle Tracking:** Oversees everything from the initial enrolment of students to their active participation in social services.
- **Unit-specific Analytics:** Generates participation summaries and activity logs for internal unit evaluation.



Unit 2
PUNSSAM88802
Sample Institute


College Dashboard

Unit Details

UNIT CODE
PUNSSAM88802

UNIT NAME
Unit 2

UNIT HEAD
Officer 3

CONTACT
9876543210

EMAIL
bsam53888@gmail.com

Quick Actions

Create Events
Manage events, volunteers and more from here.

Explore Events
Explore events created by other units.

Unit Members

Manage student volunteers

Search members...
Excel Export
UPLOAD VIA EXCEL ⓘ
+ Add Member

☐	S.NO	NAME	REG NO	DEPARTMENT	BATCH	CONTACT	ACTION
☐	1	KarthikeyanR	22CS001	CS	2022 - 2026	9876543210	✓ Edit Delete PDF
☐	2	PriyaS	22EC045	MCA	2022 - 2026	9123456789	✓ Edit Delete PDF
☐	3	DineshKumarM	22ME012	MBA	2022 - 2026	9988776655	✓ Edit Delete PDF

5. Student Enrolment & Demographic Module

- Digital Enrolment Workflow:** A comprehensive digital form system that captures student bios, academic data, and contact information.
- Demographic Analysis:** Features the specialized "Community Filtering" system to analyze volunteer diversity based on caste and community categories (OC, BC, MBC, SC, ST).
- Record Sorting:** Implements advanced sorting algorithms to organize student records by institution and unit numbers.

Add New NSS Volunteer

Basic Details

NAME * Full Name	REGISTER NUMBER * Registration Number
DEPARTMENT * e.g. CSE	COURSE * e.g. B.E. / B.Tech
BATCH FROM (YEAR) * e.g. 2022	BATCH TO (YEAR) * e.g. 2026
COMMUNITY * General	BLOOD GROUP * Select Blood Group

DATE OF BIRTH *	MOBILE PHONE NO. *
mm/dd/yyyy	10-digit mobile number

Enrolment Details

SEX *	FATHER/GUARDIAN'S NAME *
Male	Enter father/guardian's name
FATHER/GUARDIAN'S CONTACT *	HEIGHT (CM) *
10-digit mobile number	e.g. 165
WEIGHT (KG) *	AADHAAR NUMBER *
e.g. 60	12-digit Aadhaar number
EMAIL ID *	DATE OF ENROLMENT

ENTER EMAIL ADDRESS	DATE OF ENROLMENT
Enter email address	04/08/2026
CULTURAL TALENTS	HOBBIES
e.g. Singing, Dancing	e.g. Reading, Sports
UNIVERSITY NAME	
Enter full residential address	

[Download Enrolment PDF](#)
[Cancel](#)
[Add & Enrol](#)

6. Event Management & Field Activity Module

- **Outreach Documentation:** Allows coordinators to schedule events, camps, and awareness programs.
- **Multimedia Reporting:** Supports high-resolution image uploads for event galleries and detailed activity report submission for validation.

Explore Events

Discover what's happening in your unit and college

[Back to Dashboard](#)

My Unit Events

[+ Add Event](#)

Seminar
NSSEVTPUNSSAM88802001
2026-03-25

Sample Event
sample
Time: 10:00 - 17:00
Venue: Seminar Hall, Dept of Computer Science, Periyar University

Delete
Edit Report

College Events

Tree Plantation
NSSEVTPUNSSAM88801001
2026-04-08

Sample Event
sample event for testing event module
Time: 10:00 - 17:00
Venue: Periyar university Campus

Cleanliness Drive
NSSEVTPUNSSAM88801002
2026-04-09

Sample Event-3
Sample Event
Time: 10:00 - 17:00
Venue: University Campus

Sports
NSSEVTPUNSSAM88801003
2026-04-10

event-test
test
Time: 10:00 - 17:00
Venue: Ground

Workshop
NSSEVTPUNSSAM88801004
2026-04-12

mail test
hello
Time: 10:00 - 17:00
Venue: seminar hall

Other Events

Awareness Program
NSSEVTPUNSSAM33301001
2026-04-07

Sample Event-2
Sample testing event
Time: 10:00 - 15:00
Venue: Seminar Hall, Dept of Computer Science, Periyar University

7. Automated Document & Reporting Module

- **Dynamic PDF Engine:** Uses JS-driven templates to instantly generate official Government-standard Pro-forma and Enrolment forms.
- **Bulk Data Export:** Features a one-click "Export to Excel" functionality for all administrative lists, facilitating easy offline record-keeping and audit trails.
- **Bulk Data Import:** Provides a one-click "Import from Excel" feature that allows administrators to upload pre-formatted Excel sheets using a given template, enabling the addition of multiple student records at once quickly and efficiently.

NOMINATION OF NEW PROGRAMME OFFICER TO LOOK AFTER THE NSS UNIT IN THE COLLEGE SAMPLE INSTITUTE

Name: Officer 3

Designation: Professor

Department: CS

Unit Assigned:



PERSONAL DETAILS

Date of Birth: 2004-10-19

Community: OBC

Email: bsam53888@gmail.com

Mobile: 9876543210

Block: Kamalapuram

Taluk: Omalur

District: Salem

Pincode: 636309

Date of Appointment: 2020-07-19

Teaching Experience: 5

Address: 9/197 Udhayam Nagar, Kamalapuram, Omalur Salem

ACADEMIC & ETI

Educational Qualification: MCA

ETI Training Completed: Yes

Seminars / Workshops / Courses:

• NIL

GENERAL

NSS Experience:

• NIL

Special Talents / Skills:

• Dancer

ANNEXURE - A NSS VOLUNTEER ENROLMENT (AT UNIT LEVEL)

Name of the College: Sample Institute

Address:

Name of the University: Anna University

Unit No: PUNSSSAM88801

Affix passport size photo

1	Name of the Volunteer	DineshKumarM
2	Father/Guardian's Name	Marimuthu D
3	Father/Guardian's Phone No.	9977665544
4	Sex	Male [✓] Female []
5	Date of birth	2003-11-08
6	Year & Branch in which studying	B.E. - MBA
7	Community	General [] SC [] ST [] OBC []
8	Residential Address	78, Gandhi Road, Salem - 636001
9	Blood Group	A+
10	Height in 168 (cm)	Weight 70 kg
11	Mobile Phone No.	9988776655
12	E-mail ID	dinesh.m2003@gmail.com
13	Aadhaar Card No.	456789123456
14	Date/Year of Enrolment	2026-03-27
15	Cultural Talents (Please specify)	Folk Dance
16	Hobbies	Foothall, Photography

Declaration

I **DineshKumarM** hereby agree to obey all the rules and regulations of National Service Scheme and work imbuing the spirit of the Scheme.

Date: 2026-03-27

Place:

Signature of the applicant

Unit Members

Manage student volunteers

Search members...

Excel Export

UPLOAD VIA EXCEL

+ Add Member

☐	S.NO	NAME	REG NO	DEPARTMENT	BATCH	CONTACT	ACTION
☐	1	KarthikeyanR	22CS001	CS	2022 - 2026	9876543210	✓ Edit Delete PDF
☐	2	PriyaS	22EC045	MCA	2022 - 2026	9123456789	✓ Edit Delete PDF
☐	3	DineshKumarM	22ME012	MBA	2022 - 2026	9988776655	✓ Edit Delete PDF

4. TESTING AND IMPLEMENTATION

Testing is an important phase in the development of the **Periyar University NSS Portal** to ensure that the system works correctly and efficiently. It is carried out to identify and fix errors before deployment. Different types of testing are performed to validate system functionality, performance, and security. The testing process ensures that all modules operate as expected under various conditions. Overall, it helps in delivering a reliable and user-friendly application.

Functional Testing

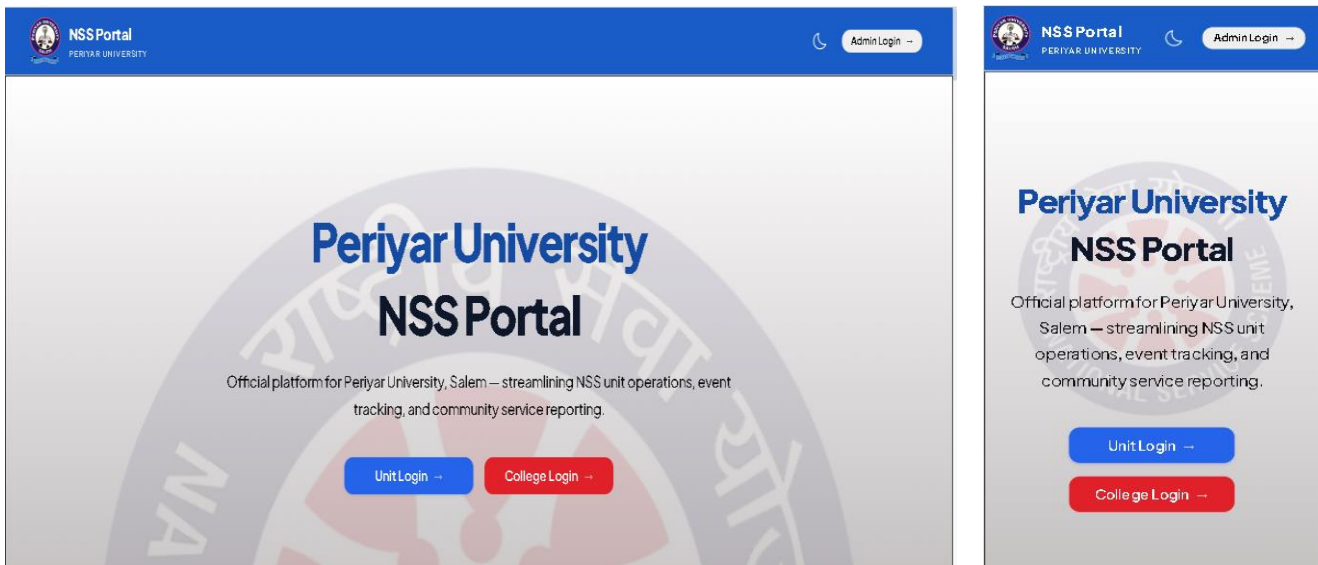
Functional testing is performed to verify that all core features of the system work as expected. The user registration process is tested to ensure unique credentials and secure password handling. Unit creation and management functionalities are validated based on defined constraints. The complete event lifecycle, including creation, updates, participation, and report submission, is thoroughly tested. This ensures that each module performs its intended function correctly.

Integration Testing

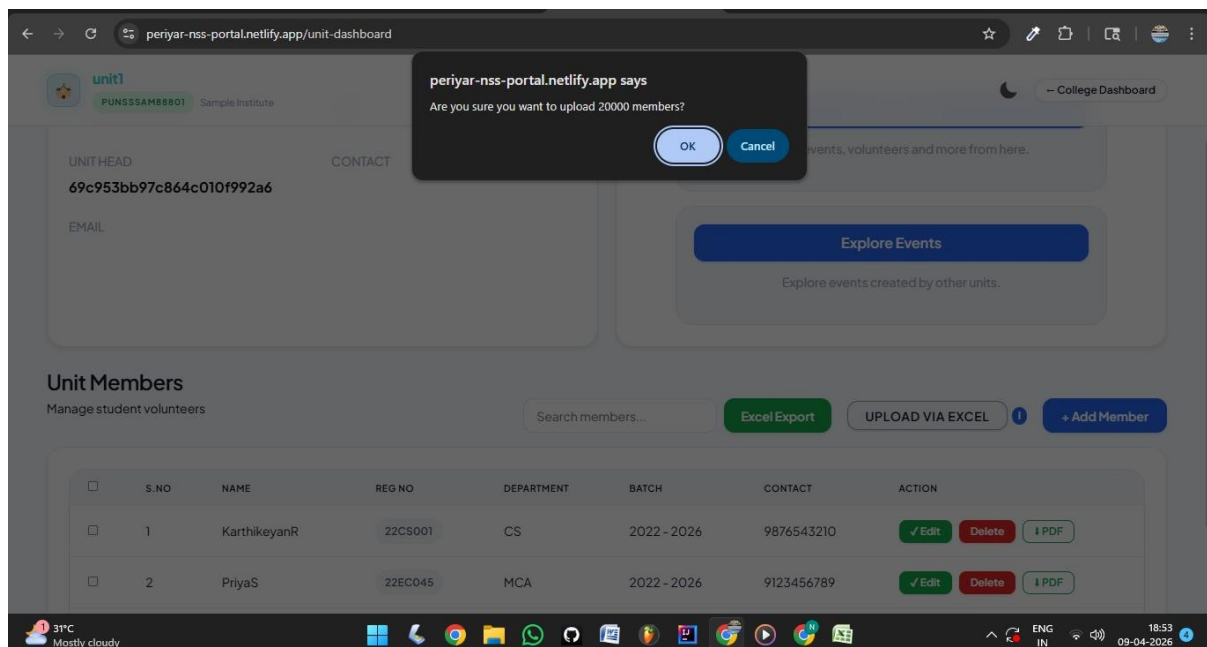
Integration testing ensures smooth interaction between different modules and external services. The communication between frontend and backend APIs is tested for accurate data transfer. Email notification services are verified to ensure alerts are sent correctly to users. Data export features such as Excel and PDF generation are tested to confirm proper formatting and data integrity. This testing guarantees that all integrated components work together seamlessly.

UI/UX Testing

UI/UX testing is conducted to ensure the system is user-friendly and visually consistent. The application is tested across different devices such as desktops, mobiles, and tablets to verify responsiveness. User interface elements like forms, buttons, and dashboards are checked for proper alignment and usability. Theme support, including light and dark modes, is also tested for readability and consistency. This improves overall user experience and accessibility.



Performance testing is carried out to evaluate the system's responsiveness and stability under different conditions. The application is tested with multiple users and large datasets to ensure smooth operation. Loading time, data processing speed, and system responsiveness are monitored. Optimizations are applied to maintain fast performance even with increased usage. This ensures reliability in real-world scenarios.



The screenshot shows a web application interface for the Periyar University NSS Portal. At the top, there is a navigation bar with the university logo, the name 'unit1', and a 'College Dashboard' link. Below this is a table listing student records. The table has columns for a checkbox, student ID, name, course code, course name, year, and contact number. Each row includes 'Edit', 'Delete', and 'PDF' action buttons. At the bottom, a pagination bar shows 'Showing 20001 to 20013 of 20013 students' and a set of page numbers from 397 to 401, with '401' being the active page.

	ID	Name	Course Code	Course Name	Year	Contact	Actions
☐	20006	DineshKumarM	22ME012	MBA	2022 - 2026	9988776655	✓ Edit Delete PDF
☐	20007	Sarath	22CS001	CS	2022 - 2026	9876543210	✓ Edit Delete PDF
☐	20008	Sam	22EC045	MCA	2022 - 2026	9123456789	✓ Edit Delete PDF
☐	20009	Deepak	22ME012	MBA	2022 - 2026	9988776655	✓ Edit Delete PDF
☐	20010	Gowtham	22CS008	CS	2022 - 2026	9876543210	✓ Edit Delete PDF
☐	20011	Mohammed	22EC049	MCA	2022 - 2026	9123456789	✓ Edit Delete PDF
☐	20012	Vinoth	22ME030	MBA	2022 - 2026	9988776655	✓ Edit Delete PDF
☐	20013	Vignesh	22ME048	MBA	2022 - 2026	9988776655	✓ Edit Delete PDF

Showing 20001 to 20013 of 20013 students

Previous 397 398 399 400 **401** Next

The system was evaluated by importing a large dataset of around 20,000 student records at once. Although the data was successfully uploaded, a performance issue occurred while displaying the student information due to the large volume of data.

To resolve this issue, pagination was implemented, which limits the number of records displayed per page. This significantly improved performance and ensured smooth data retrieval and user experience. The solution effectively handled large-scale data without affecting system efficiency.

Security Testing

Security testing is implemented to protect user data and system integrity. Authentication mechanisms are tested to ensure only authorized users can access the system. Token-based security is verified for protected routes and sensitive operations. Data encryption techniques are checked to safeguard user credentials. This ensures the system remains secure against unauthorized access and threats.

IMPLEMENTATION AND DEPLOYMENT

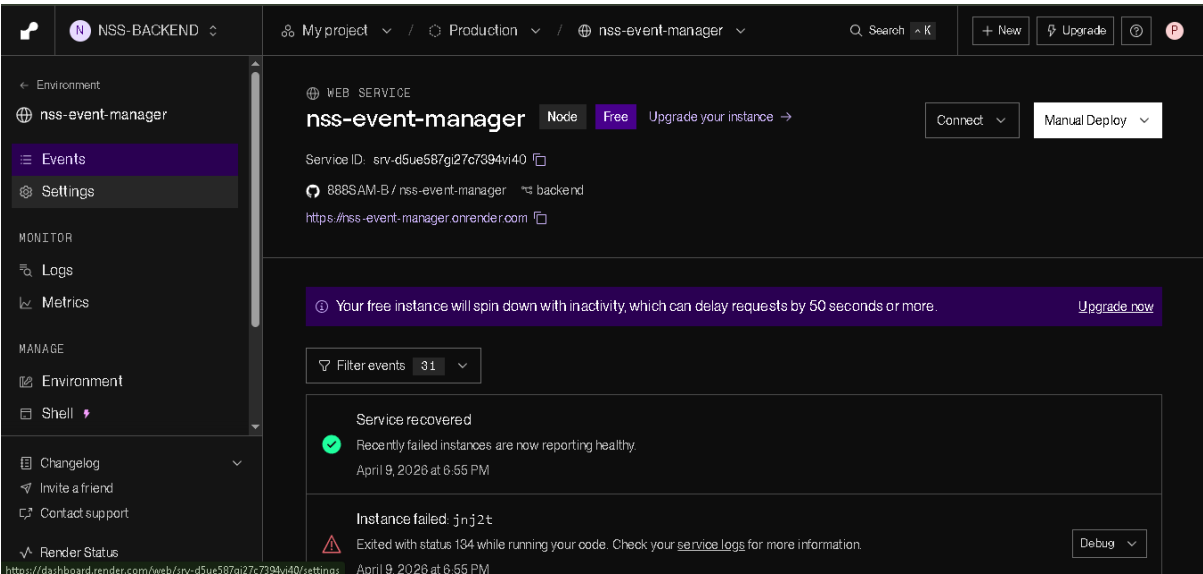
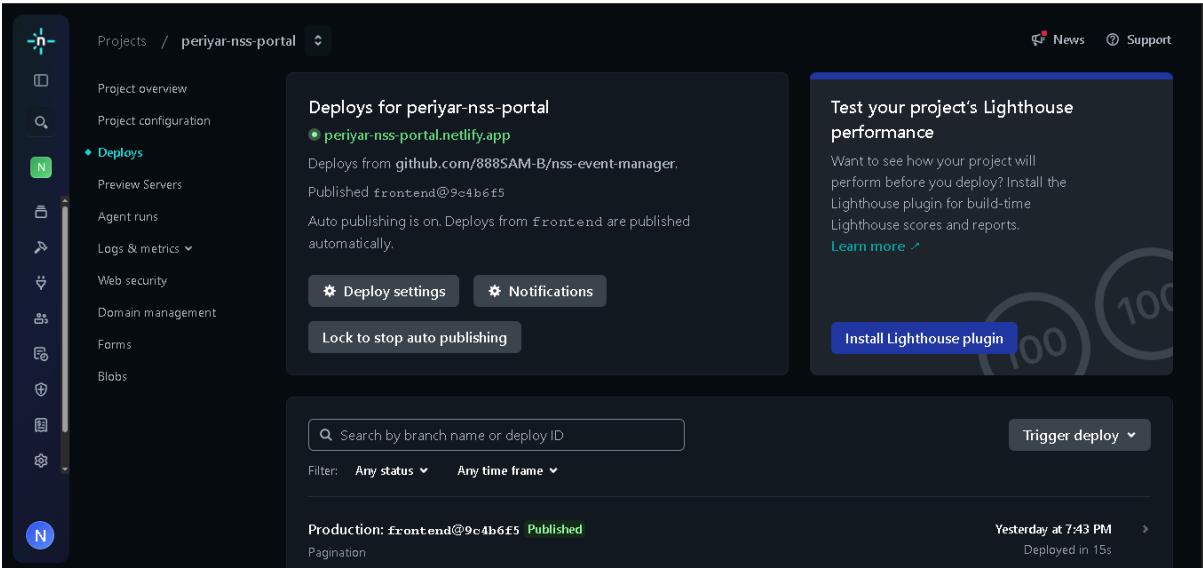
IMPLEMENTATION PHASE

The implementation phase involves the development and integration of all modules of the **Periyar University NSS Portal**. During this phase, the system is built using a structured approach where frontend, backend, and database components are developed and connected. All

features such as user authentication, volunteer management, event handling, and reporting are implemented based on the system design. Proper coding standards and validation techniques are followed to ensure accuracy and efficiency. Continuous testing is also carried out during this phase to identify and fix errors.

DEPLOYMENT PHASE

The Periyar University NSS Portal was deployed using modern cloud platforms to ensure accessibility and reliability. The frontend was hosted on Netlify, while the backend services were deployed using Render for efficient server-side operations. The database was managed using MongoDB Atlas, ensuring secure and scalable data storage. All components were deployed directly from GitHub, enabling continuous integration and easy updates. This approach ensures high availability, performance, and simplified maintenance.



5.CONCLUSION

The **Periyar University NSS Portal** is a comprehensive web-based solution designed to modernize and streamline the management of National Service Scheme activities. It replaces traditional manual and paper-based processes with a centralized digital platform, enabling efficient handling of volunteer enrollment, event management, attendance tracking, and report generation. This significantly reduces manual effort, minimizes errors, and improves overall data accuracy and management.

The system follows a role-based access structure, allowing Admin, College, and Unit users to perform their responsibilities in a secure and organized manner. It is structured into key modules such as Authentication & Secure Access, Administrator Command, College Administration, Unit Operations & Volunteer Management, Student Enrolment & Demographic, Automated Document & Reporting, and Event Management & Field Activity, ensuring efficient system functionality.

The application is developed using modern technologies such as React for the frontend, Node.js and Express for the backend, and MongoDB Atlas for database management, providing a scalable and high-performance environment. The platform also enhances communication through automated notifications and offers a user-friendly interface with structured dashboards.

Overall, the system improves transparency, coordination, and efficiency, making it a reliable and future-ready solution for managing NSS activities.

6.BIBLIOGRAPHY

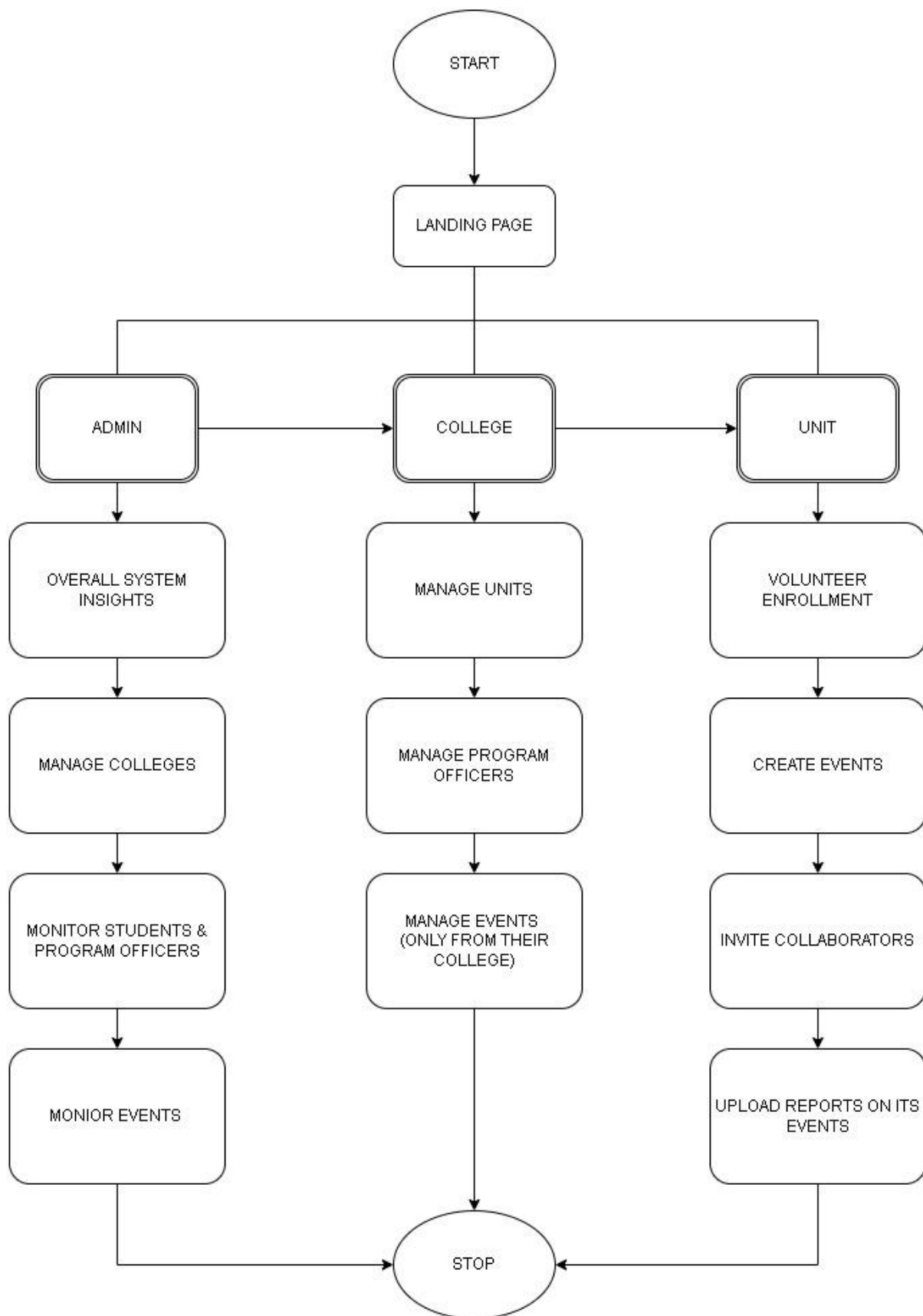
Web References

- Official MongoDB Documentation – <https://www.mongodb.com/docs/>
- React Documentation – <https://react.dev/>
- Node.js Documentation – <https://nodejs.org/en/docs/>
- Express.js Documentation – <https://expressjs.com/>
- Mongoose Documentation – <https://mongoosejs.com/docs/>
- Chart.js Documentation – <https://www.chartjs.org/docs/>
- Netlify Documentation – <https://docs.netlify.com/>
- Render Documentation – <https://render.com/docs/>
- MongoDB Atlas Documentation – <https://www.mongodb.com/atlas/docs/>
- GitHub Documentation – <https://docs.github.com/>

Book References

- Web Development with Node and Express – Ethan Brown
- Learning React – Alex Banks & Eve Porcello
- MongoDB: The Definitive Guide – Kristina Chodorow
- JavaScript: The Good Parts – Douglas Crockford
- Eloquent JavaScript – Marijn Haverbeke

A.DATA FLOW DIAGRAM



B.TABLE STRUCTURE

MEMBER TABLE

Field Name	Type	Description
name	String	Member name
regNo	String	Register number
dept	String	Department
course	String	Course name
community	String	Community category
contact	String	Phone number
bloodGroup	String	Blood group
dob	String	Date of birth
batchFrom	String	Batch start year
batchTo	String	Batch end year
unitId	ObjectId (Unit)	Reference to Unit
collegeld	ObjectId (User)	Reference to College
fatherName	String	Father name
fatherPhone	String	Father contact
sex	String	Gender
height	String	Height
weight	String	Weight
aadhaar	String	Aadhaar number
enrolmentDate	String	NSS enrolment date
culturalTalents	String	Talents
hobbies	String	Hobbies
address	String	Address
enrolmentNo	String	NSS enrolment number
remarks	String	Remarks
universityName	String	University name
email	String	Email
isEnrolled	Boolean	Enrolment status

UNIT TABLE

Field Name	Type	Description
name	String	Unit name
head	Mixed (ProgramOfficer)	Unit head
contact	String	Contact number
mail	String	Email
password	String	Hashed password
members	ObjectId[] (Member)	List of members
unitNumber	String	NSS unit number
createdDate	String	Created date
events	Array	Events list
collegeId	ObjectId (User)	College reference

USER (COLLEGE) TABLE

Field Name	Type	Description
userName	String	Login username
password	String	Hashed password
insName	String	Institution name
location	String	Location
block	String	Block
taluk	String	Taluk
district	String	District
pincode	String	Pincode
universityName	String	University
events	Array	Events
code	String	Unique code
units	ObjectId[] (Unit)	Units under college
adoptingVillages	Array	Village details

PROGRAM OFFICER TABLE

Field Name	Type	Description
name	String	Officer name
designation	String	Designation
department	String	Department
unit	String	Unit
college	String	College
image	String	Photo
dob	String	DOB
community	String	Category
email	String	Email
mobile	String	Phone
address	String	Address
block	String	Block
taluk	String	Taluk
district	String	District
pincode	String	Pincode
dateOfAppointment	String	Appointment date
teachingExperience	String	Experience
qualification	String	Qualification
seminars	String[]	Seminars
etiCompleted	String	ETI completed
nssExperience	String[]	NSS experience
specialTalent	String[]	Talents
collegeId	ObjectId (User)	College reference
officerID	String	Unique ID
createdAt	Date	Created date

EVENT TABLE

Field Name	Type	Description
name	String	Event name
description	String	Description
category	String	Event type
singleDay	Boolean	Single/multi-day
date	String	Single day date
dateFrom	String	Start date
dateTo	String	End date
timeFrom	String	Start time
timeTo	String	End time
venue	String	Venue
images	Array	Image URLs
brochure	String	Brochure file
eventCode	String	Unique code
unitId	ObjectId (Unit)	Unit reference
attendees	ObjectId[] (Member)	Participants
collegeId	ObjectId (User)	College reference

Report Sub-Table

Field Name	Type
conductedOnDate	Boolean
participantsCount	Number
collegesCount	Number
outcome	String
reportFile	String
reportPhotos	Array
submittedAt	Date

Adopting Villages Sub-Table

Field Name	Type
name	String
address	String
block	String
taluk	String
district	String
pincode	String

C.SAMPLE CODING

```
const express = require('express');

const mongoose = require('mongoose');

require('dotenv').config();

const jwt = require('jsonwebtoken');

const cors = require("cors");

const bcrypt = require('bcryptjs');

const verifyToken = require('./middleware/auth');

if (process.env.BREVO_API_KEY) {

  console.log(' Brevo Email service is configured');

} else {

  console.warn(' BREVO_API_KEY is not set in environment variables');

}


const sendEmailViaBrevo = async ({ from, to, subject, html, bcc }) => {

  const formatEmail = (email) => {

    if (typeof email === 'object') return email;

    return { email };

  };

  const toArr = Array.isArray(to) ? to.map(formatEmail) : (to ? [{ email: to }] : []);
```

```

const bccArr = Array.isArray(bcc) ? bcc.map(formatEmail) : (bcc ? [{ email: bcc }] : []);

const payload = {
  sender: formatEmail(from || process.env.EMAIL_USER || 'noreply@example.com'),
  subject: subject,
  htmlContent: html,
};

if (toArr.length > 0) payload.to = toArr;

if (toArr.length === 0 && bccArr.length > 0) {
  payload.to = [formatEmail(from || process.env.EMAIL_USER ||
'noreply@example.com')];
}

if (bccArr.length > 0) payload.bcc = bccArr;

const response = await fetch('https://api.brevo.com/v3/smtp/email', {
  method: 'POST',
  headers: {
    'Accept': 'application/json',
    'api-key': process.env.BREVO_API_KEY,

```

```

        'Content-Type': 'application/json'

    },

    body: JSON.stringify(payload)

});

if (!response.ok) {

    const contentType = response.headers.get('content-type');

    let errData = await response.text();

    if (contentType && contentType.includes('application/json')) {

        try { errData = JSON.parse(errData); } catch (e) { }

    }

    const error = new Error(`Brevo API Error: ${typeof errData === 'object' ?
JSON.stringify(errData) : errData}`);

    error.code = response.status;

    throw error;

}

const data = await response.json();

return { response: '250 Message queued', messageId: data.messageId || 'unknown' };

app.listen(PORT, () => {

    console.log(`Server is running on http://localhost:${PORT}`);

});

```


D.SAMPLE OUTPUT



Sample Institute

888 NSS Organization

Explore Events

Admin

Adopting Villages

1 Village(s) + Add Village

R.C.Chettipatty

In between omalur and kamalapuram

Block: OmalurTaluk: OmalurDist: Salem

636309

NSS Units Management

Manage your college NSS units and members

View Program Officers

View All Students

+ Create New Unit

Register Program Officer

unit1

PUNSSSAM88801

Head: officer 3

Created: 2026-03-27

Members: 20013

Delete Unit

Unit 2

PUNSSSAM88802

Head: Officer 3

Created: 2026-03-29

Members: 6

Delete Unit

unit1

PUNSSSAM88801

Sample Institute

College Dashboard

Unit Details

UNIT CODE

PUNSSSAM88801

UNIT NAME

unit1

UNIT HEAD

officer 3

CONTACT

9876543210

EMAIL

catchyourcourse@gmail.com

Quick Actions

Create Events

Manage events, volunteers and more from here.

Explore Events

Explore events created by other units.

Unit Members

Manage student volunteers

Search members...

Excel Export

UPLOAD VIA EXCEL

+ Add Member

☐	S.NO	NAME	REG NO	DEPARTMENT	BATCH	CONTACT	ACTION
☐	1	KarthikeyanR	22CS001	CS	2022 - 2026	9876543210	✓ Edit Delete ↓ PDF
☐	2	PriyaS	22EC045	MCA	2022 - 2026	9123456789	✓ Edit Delete ↓ PDF
☐	3	DineshKumarM	22ME012	MBA	2022 - 2026	9988776655	✓ Edit Delete ↓ PDF

33